

ETW2001 Foundations of Data Analysis
SEM 1 2026
Assignment 2
TOTAL MARKS (40%)

Objective

The objective of this assignment is to design an interactive dashboard using the Superstore dataset along with additional external dataset(s), and to communicate insights through visualizations and a structured data story.

Tasks

1. Dataset Requirement

- Use the **Superstore dataset** that was explored in Assignment 1.
- Identify and include **at least one additional external dataset**.
- The external dataset must be:
 - Relevant to your chosen problem
 - From a credible source (e.g., government, academic, or recognized data platform)

2. Dataset Selection and Justification

- Clearly cite the source of the external dataset.
- Provide a justification for selecting this dataset.
- Explain how it complements the Superstore dataset.
- Include a brief description of the external dataset

3. Problem Design

- Define a clear problem statement.

4. R Shiny Dashboard Development

Create an **interactive dashboard** using R Shiny that includes:

- A clean and structured layout
- At least **5 different visualizations**
- Interactive features such as:
 - Filters (dropdowns, sliders, checkboxes, etc.)
 - Dynamic updates of charts based on user input
- The dashboard should:
 - Be user-friendly

- Allow exploration of the data
- Clearly present information relevant to your problem

5. Visualization and Interpretation

- For each of the 5 visualizations:
 - Provide a clear interpretation
 - Explain what the visualization shows
 - Explain how it contributes to answering the problem

6. Data Story Development

- Combine all findings into a **coherent data story**.
- The data story should:
 - Be written in a narrative format
 - Connect all visual insights together
 - Explain the overall conclusion and implications

Report Structure (10 Pages)

Page 1: Dashboard Screenshot

- Include a full screenshot of your R Shiny dashboard

Page 2: Dataset Selection

- Superstore dataset (brief mention)
- External dataset:
 - Citation
 - Justification
 - Description of the dataset

Pages 3-7: Visualizations and Interpretation

- One visualization per page
- Each page must include:
 - Chart (top half)
 - Interpretation (bottom half)

Pages 8-9: Data Story

- Present the complete data narrative
- Summarize insights derived from all visualizations

Page 10: AI Acknowledgement and Citations

- Include:
 - AI usage acknowledgement (if applicable)
 - All other citations and references

Submission Requirements

Submit the following files:

1. R Shiny app file (`app.R`)
2. External dataset files
3. Any processed/cleaned datasets used for dashboard design
4. Final 10-page PDF report

Criteria	HD (High Distinction) 5	D (Distinction) 4	C (Credit) 3	P (Pass) 1	N (Fail) 0	Marks
Problem Design & Objective	Exceptionally clear, well-defined, highly relevant problem	Clear and relevant problem with well-stated objectives	Adequate problem and somewhat clear objectives	Basic problem stated, lacks clarity	Unclear or missing problem	/5
Dataset Selection, Citation & Justification	Uses 3 external datasets , all well-justified and correctly cited	Uses 2 external datasets , justified and cited	Uses 1 external dataset , adequately justified and cited	Uses 1 external dataset with weak justification or citation	No external dataset used	/5
Dashboard Design & Interactivity	Highly polished, intuitive, excellent interactivity and usability	Well-structured with good interactivity	Functional with basic interactivity	Limited interactivity, cluttered design	Poor or non-functional dashboard	/5
Visualizations Quality (5 charts)	5 different types of charts , includes information from all selected datasets	5 different types of charts , includes information from at least one external dataset	5 charts but lack clarity or effectiveness	Repeated chart types or poor design	Less than 5 charts and/or repeated types	/5
Interpretation of Visualizations	Insightful, clear, well-explained interpretations	Good interpretations with clear explanations	Basic explanations with some gaps	Limited or superficial interpretation	Missing or incorrect interpretations	/10
Data Story Narrative	Strong, coherent, compelling narrative connecting findings	Clear and logical narrative	Adequate narrative with some inconsistencies	Weak or disjointed narrative	No clear data story	/10

TOTAL MARKS

40